

Modified Booth Algorithm

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Modified Booth's Algorithm

- Guarantees that the maximum number of summands that must be added is $n/2$ for n -bit operands.
- Bit pair recoding technique
- Observe the following:
 - The pair $(+1, -1)$ is equivalent to the pair $(0, +1)$
 - $(+1, 0)$ is equivalent to $(0, +2)$
 - $(-1, +1)$ is equivalent to $(0, -1)$
 - $(-1, 0)$ is equivalent to $(0, -2)$
 - Examines 3 bits at a time.

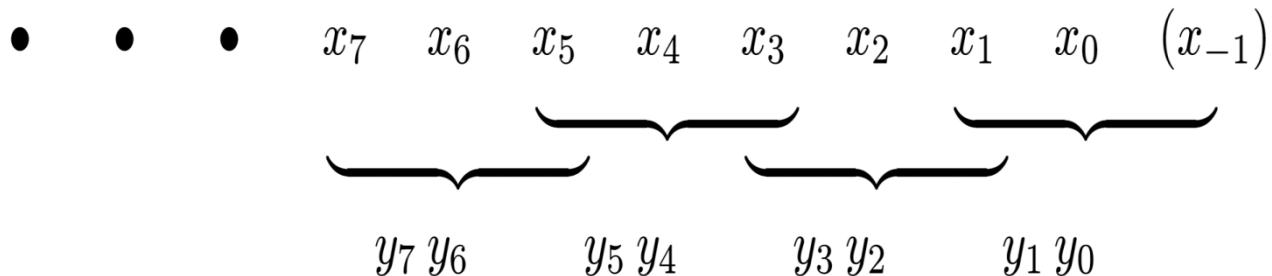


Table of Multiplicand and Selection decisions

Multiplier Bit-Pair		Multiplier bit on the right $i - 1$	Booth Multiplier bits		Multiplicand selected at position i
$i + 1$	i				
0	0	0	0	0	$0 \times M$
0	0	1	0	1	$+ 1 \times M$
0	1	0	1	-1	$+1 \times M$
0	1	1	1	0	$+ 2 \times M$
1	0	0	-1	0	$- 2 \times M$
1	0	1	-1	1	$- 1 \times M$
1	1	0	0	-1	$-1 \times M$
1	1	1	0	0	$0 \times M$

$$\begin{array}{r}
 0 \ 1 \ 1 \ 0 \ 1 \ (+13) \\
 \times 1 \ 1 \ 0 \ 1 \ 0 \ (-6) \\
 \hline
 \end{array}$$



Multiplication requiring only $n/2$ summands

$$\begin{array}{r}
 \begin{array}{cccccc} 0 & 1 & 1 & 0 & 1 & \end{array} \\
 \begin{array}{cccccc} 0 & -1 & +1 & -1 & 0 & \end{array} \\
 \hline
 \begin{array}{cccccccccc} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 & 1 & 0 & 0 & 1 & 1 & \\ 0 & 0 & 0 & 0 & 1 & 1 & 0 & 1 & & \\ 1 & 1 & 1 & 0 & 0 & 1 & 1 & & & \\ 0 & 0 & 0 & 0 & 0 & 0 & & & & \end{array} \\
 \hline
 \begin{array}{cccccccccc} 1 & 1 & 1 & 0 & 1 & 1 & 0 & 0 & 1 & 0 \end{array} \quad (-78)
 \end{array}$$



$$\begin{array}{r}
 \begin{array}{cccccc} 0 & 1 & 1 & 0 & 1 & \end{array} \\
 \begin{array}{cccccc} 0 & & -1 & & -2 & \end{array} \\
 \hline
 \begin{array}{cccccccccc} 1 & 1 & 1 & 1 & 1 & 0 & 0 & 1 & 1 & 0 \\ 1 & 1 & 1 & 1 & 0 & 0 & 1 & 1 & & \\ 0 & 0 & 0 & 0 & 0 & 0 & & & & \end{array} \\
 \hline
 \begin{array}{cccccccccc} 1 & 1 & 1 & 0 & 1 & 1 & 0 & 0 & 1 & 0 \end{array}
 \end{array}$$

Modified Booth Algorithm

1. M = Multiplicand (n bits),
 $A = 0$, Q = Multiplier (n bits)
2. $\text{Count} = n/2$
3. While ($\text{Count} \neq 0$)
 - a. If $Q_1Q_0Q_{-1} = 001$
 - i. $A \leftarrow A + M$
 - b. Else if $Q_1Q_0Q_{-1} = 010$
 - i. $A \leftarrow A + M$
 - c. Else if $Q_1Q_0Q_{-1} = 011$
 - i. shift left M by 1 bit
 - ii. $A \leftarrow A + M$
 - d. Else if $Q_1Q_0Q_{-1} = 100$
 - i. shift left M by 1 bit
 - ii. $A \leftarrow A + \overline{M} + 1$
 - e. Else if $Q_1Q_0Q_{-1} = 101$
 - i. $A \leftarrow A + \overline{M} + 1$
 - f. Else if $Q_1Q_0Q_{-1} = 110$
 - i. $A \leftarrow A + \overline{M} + 1$
 - g. Arithmetic shift right A, Q, Q_{-1}
 - h. Arithmetic shift right A, Q, Q_{-1}
 - i. $\text{Count} \leftarrow \text{Count} - 1$

Modified Booth (-9x-13)

$M = -9 = 110111$, $Q = -13 = 110011$, $SC = 3$

Comment	A	Q	Q _{n+1}	SC
Initially	000000	110011	0	3
Subtract M (-1xM)	001001			
	001001			
ashr	000100	111001	1	
ashr	000010	011100	1	2
Add M (+1xM)	110111			
	111001			
ashr	111100	101110	0	
ashr	111110	010111	0	1
Subtract M (-1xM)	001001			
	000111			
ashr	000011	101011	1	
ashr	000001	110101	1	0

Modified Booth (13 x -6)

$M = 13 = 001101$, $Q = -6 = 111010$, $SC = 3$

Comment	A	Q	Q _{n+1}	SC
Initially	000000	111010	0	3
(-2xM), shl M and 2's complement	100110			
	100110			
Ashr (2 times)	111001	101110	1	2
(-1xM) subtract M	110011			
	101100			
Ashr (2 times)	111011	001011	1	1
Ashr (2 times)	111110	110010	1	0

$13 \times -6 = -78 = 111110110010$